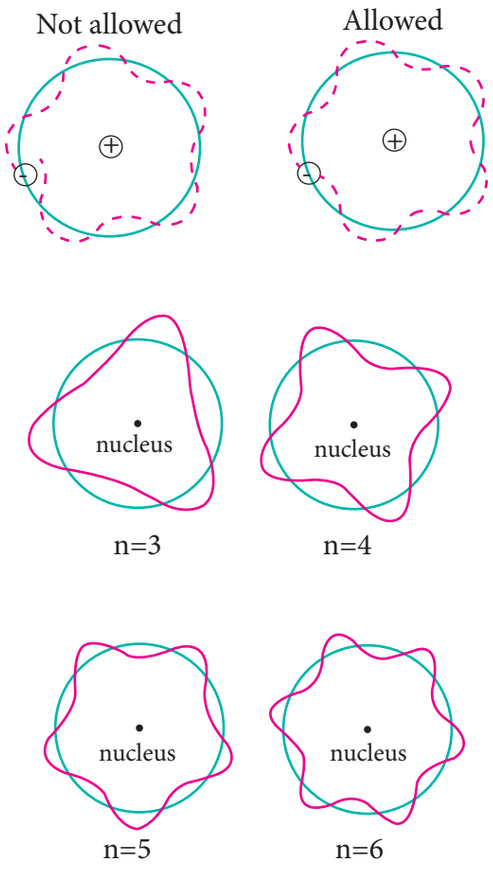

Figure. 2.2 Wave nature of electrons in allowed Bohr orbits

Davison and Germer experiment :

The wave nature of electron was experimentally confirmed by Davisson and Germer. They allowed the accelerated beam of electrons to fall on a nickel crystal and recorded the diffraction pattern. The resultant diffraction pattern is similar to the x-ray diffraction pattern. The finding of wave nature of electron leads to the development of various experimental techniques such as electron microscope, low energy electron diffraction etc…

2.3 Heisenberg's uncertainty principle

The dual nature of matter imposes a limitation on the simultaneous determination of position and momentum of a microscopic particle. Based on this, Heisenberg arrived at his uncertainty principle, which states that 'It is impossible to **accurately** determine both the position and the momentum of a microscopic particle **simultaneously**'. The product of uncertainty (error) in the measurement is expressed as follows.

$$\Delta x.\Delta p \geq h/4\pi \text{ -------- } (2.11)$$

where, Δx and Δp are uncertainties in determining the position and momentum, respectively.

The uncertainty principle has negligible effect for macroscopic objects and becomes significant only for microscopic particles such as electrons. Let us understand this by calculating the uncertainty in the velocity of the electron in hydrogen atom. (Bohr radius of 1st orbit is 0.529 Å) Assuming that the position of the electron in this orbit is determined with the accuracy of 0.5 % of the radius.

Uncertainity in position = Δx

$$= \frac{0.5\,\%}{100\,\%} \text{ x } 0.529 \text{ Å}$$

$$= \frac{0.5}{100} \text{ x } 0.529 \text{ x } 10^{-10} \text{ m}$$

$$\Delta x = 2.645 \text{ x } 10^{-13} \text{ m}$$

From the Heisenberg's uncertainity principle,

$$\Delta x.\Delta p \geq \frac{h}{4\pi}$$

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